# THE HENSELIZATION AND STRICT HENSELIZATION MAPS ARE FORMALLY ETALE AND FORMALLY SMOOTH

THE STACKS PROJECT AUTHORS

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*Allowed standard prerequisites (editorial).*

- Etale algebras are formally etale, and colimits of formally etale algebras are formally etale. Algebra, Lemma \ref{algebra-lemma-formally-etale-etale} and Algebra, Lemma \ref{algebra-lemma-colimit-formally-etale}, cited at proof lines 10969-10972. Give formal etaleness of the three maps.
- A formally etale ring map is formally smooth, and formal smoothness upgrades to the adic topology. Algebra, the implication from formal etaleness to formal smoothness, and Lemma \ref{lemma-formally-smooth}, cited at proof lines 10973-10976. Give the second part of the statement.

*Source reference.* Original lemma 07QN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9838–9838. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

*Source reference.* Original lemma 00UR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1121–1121. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 031N: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3990–3990. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

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*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 07EC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9442–9442. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

**Lemma 0.2.** *Let $(R, \mathfrak{m}, \kappa)$ be a local ring. Then*

(1) *$R \to R^h$, $R^h \to R^{sh}$, and $R \to R^{sh}$ are formally étale,*
(2) *$R \to R^h$, $R^h \to R^{sh}$, resp. $R \to R^{sh}$ are formally smooth in the $\mathfrak{m}^h$, $\mathfrak{m}^{sh}$, resp. $\mathfrak{m}^{sh}$-topology.*

*Proof.* Part (1) follows from the fact that $R^h$ and $R^{sh}$ are directed colimits of étale algebras (by construction), that étale algebras are formally étale (Algebra, Lemma 00UR), and that colimits of formally étale algebras are formally étale (Algebra, Lemma 031N). Part (2) follows from the fact that a formally étale ring map is formally smooth and Lemma 07EC. □